HACK MATRIX HACKATHON

TITLE PAGE

- PROJECT TITLE : E-WASTE DISPOSAL GUIDE

- TEAM NAME : THE CORE FOUR

- TEAM MEMBERS : SHATABDI SINGH (24BCE10912)
  SAMRIDHI TYAGI (24MIM10091)
  ASHUTOSH RAI (24BCE11359)
  SALONI TEOTIA (25MIM10155)

PROBLEM STATEMENT

- **PROBLEM :-**

Every day, people throw away old electronics in regular dustbins. This happens mainly because they don't know how or where to dispose of them safely. E-waste contains hazardous materials such as lead, mercury and lithium. The core issue is a lack of awareness and lack of accessible information.. This problem is about bridging the gap between people's intention to dispose of electronicss responsibly and the lack of simple, accessible information/tools to do it.

Who is affected by this problem

- General Public
- Urban Communities and Cities
- Environment (environment and ecosystems)
- Waste Workers

Why this problem is important

- Toxic chemicals released from e-waste can remain in the environment for years, contributing to climate and pollution issues.
- India loses ₹7,000+ crore in recoverable metals every year, high repair costs push people to buy new devices → resource wastage.

EXISTING SOLUTIONS AND GAPS

➢ Limitations

- Dependence on accurate location data
- Limited availability of e-waste centers
- Internet dependency
- Limited scope of solution

➢ Our solution targets the gap by:

- Letting users search for a device (eg; old laptop, old phone)
- Showing safe disposal methods.
- Displaying nearby authorized drop-off locations on a map.

PROPOSED SOLUTION

SudhaarSetu is a web-based platform that connects consumers, authorized repair shops, and recycling centers to reduce e-waste and promote sustainable electronics usage.

- **Key Features**

1. Repair Cost Estimator (AI-powered)
2. Repair Hub
3. Digital Repair Passport(Device history log)
4. SudharBot (Chatbot)
5. Repair & Recycling Locator
6. E-Waste Collection Booking
7. Incentive & Credit System

Explain how it solves the problem

1. Addresses the root cause: lack of awareness
2. Provides easy access to disposal locations.
3. Goes beyond disposal, expands the solution: repair option, reuse option
4. Promotes sustainable behavior

SYSTEM ARCHITECTURE

USER LOGIN & REGISTERATION SYSTEM

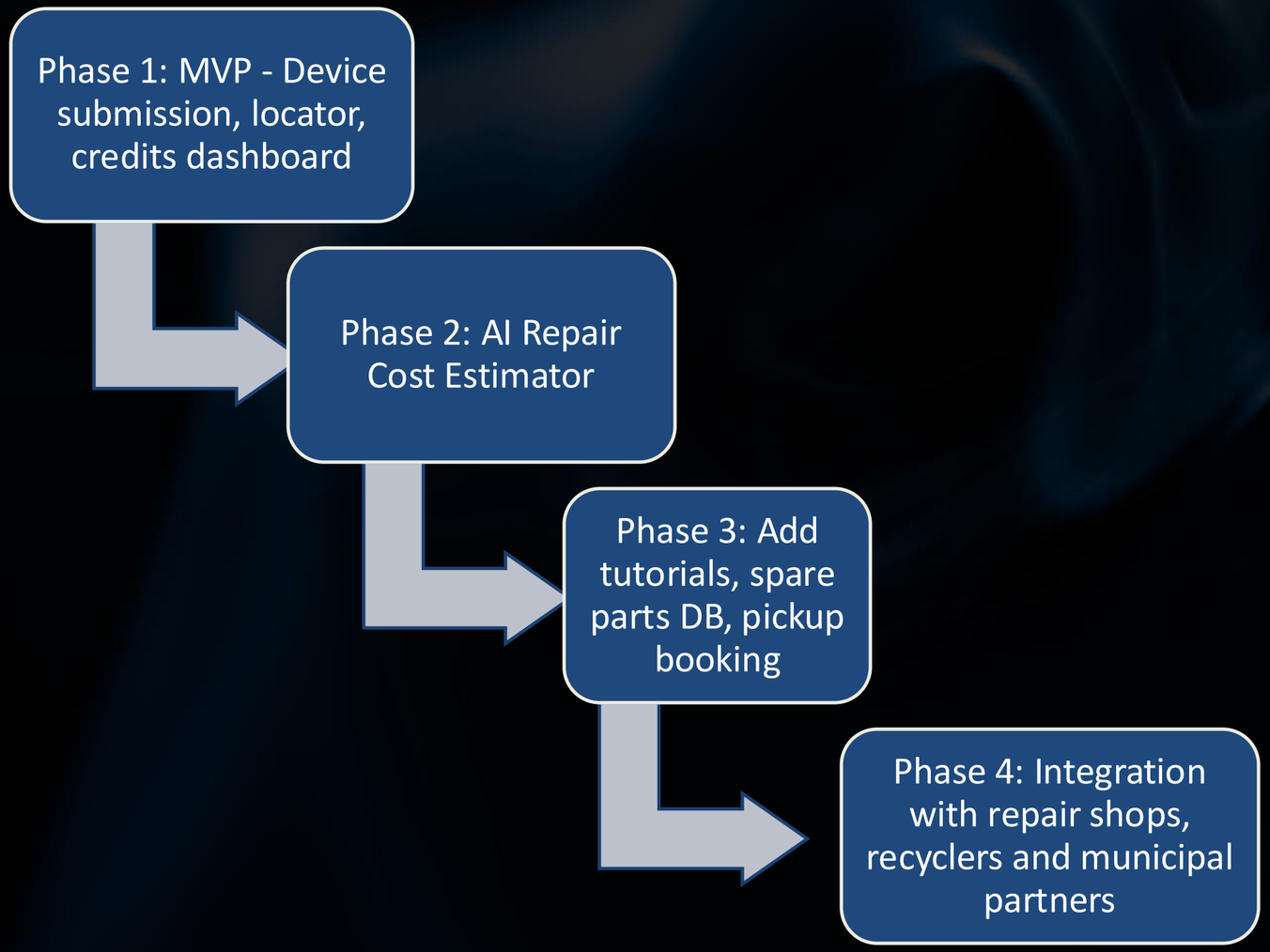
FRONTEND
(React Interface)
Booking UI, Pickup UI, Sudhaar Bot

BACKEND
(Node.js + Express)
(API Routes, Auth Handling)

DATABASE
(MongoDB)
(Users, Bookings, Repair History

FINAL TESTING & INTEGRATION

FLOWCHART / WORKING PROCESS



TECH STACK

FRONTEND

HTML, CSS, JavaScript

BACKEND

MongoDB – Database(JSON-like flexible storage)
React.js – Web
Node.js – Server Runtime

MAPS & LOCATION

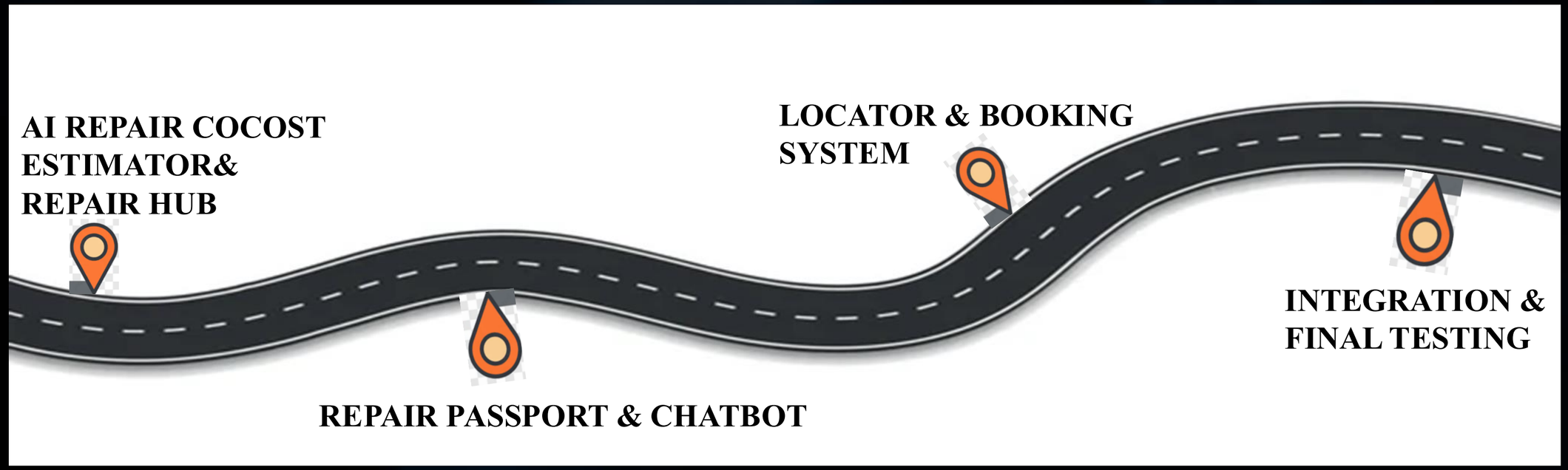
Google Maps API

AUTHENTICATION

Phone-based login

Month 1–2: UI/UX design, core backend setup, database schema, user authPhase 2 — Core FeaturesMonth 3–4: AI Estimator, repair passport, chatbot, locator, ecosystemMonth 5–6: Shop onboarding, e-waste booking, incentive system, payment gatewayPhase 4 — LaunchMonth 7–8: Beta testing, feedback

IMPLEMENTATION PLAN



EXPECTED OUTCOME

- **Benefits**
  - Easy access to correct disposal information.
  - Encourages repair instead of replacement.
  - More inflow of e-waste through proper channels.
  - Saves time searching for recycling centers

- **Real-world impact**
  - Lower pollution of soil, water and air.
  - Reduction in improper e-waste disposal.
  - Supports government and sustainability initiatives.
  - Reduces need for raw material extraction.

- **Expected results**
  - Device search system: users can search for any electronic item
  - Disposal & repair guidance: step-by-step instructions
  - Interactive map: Displays nearby recycling centers, authorized repair shops.

FUTURE SCOPE AND CONCLUSION

- **Summary**

SudhaarSetu is a user-friendly web platform designed to tackle the growing the issue of E-waste by connecting consumers with authorized repair shops and recycling centers, contributing to a cleaner and more sustainable environment.

It simplifies the process of:

- Finding safe disposal methods.
- Locating nearby service providers.
- Encouraging repair and reuse instead of disposal.

- **Future improvements**

- Suggest the best option for: repair, reuse, or recycle.
- Smart tips based on device condition.
- Ensure verified recycling centers.
- Improves usability in low-network areas.